NCS2K USB GUI User Manual

V1.02

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Document History

Rev	Date	Author	Comment
1.00	2019-01-24	Zhidong Ruan	1. Initial release
1.01	2019-01-31	Zhidong Ruan	1. Add the method of upgrade firmware from 4912 to 491B 2. Add the method of upgrade Bin Data.
1.02	2019-04-01	Zhidong Ruan	1. Add the function of refresh IL data in PCAMAP 2. Update NCS2K_V1.0.0.2.exe to NCS2K_V1.0.0.3.exe

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Doc Approved by	Xin Lu

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Table of Contents

1 Purpose 4

2 Environment and Configuration of GUI 4

3 LIBUSBK Installation Tutorial 5

4 GUI Instructions 11

4.1 Connect Module 11
4.2 Command Executing 12
4.3 Firmware Upgrade 13
4.4 Upgrade firmware to change Equipment ID (4912 to 491B) 14
4.5 How to Upgrade PCAMAP bin file 15

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1 Purpose

This document mainly describes the NCS2K product GUI user manual, LIBUSBK driver installation tutorial. Through this GUI, product users can communicate with NCS2K products.

2 Environment and Configuration of GUI

To communicate with the product properly, you need to configure the GUI's usage environment. Table 2.1 shows the Environment configuration. Before use this GUI, LIBUSBK must be installed first. Section 3 shows the the LIBUSBK Installation Tutorial and section 4 is the manual of GUI.

Table 2.1： Environment configuration

Environment	Default Setting
Application System	Windows 10
USB Drive	LIBUSBK 3.0.7.0
Product module	NCS2K-MF-DEG-5-CV VID:0x05ae PID:490E NCS2K-MF-UPG-4-CV VID:0x05ae PID: 490F NCS2K-MF-M16LC-CV VID:0x05ae PID: 4912 NCS2K-MF-8X10G-FO VID:0x05ae PID: 491B
USB cable	USB 2.0

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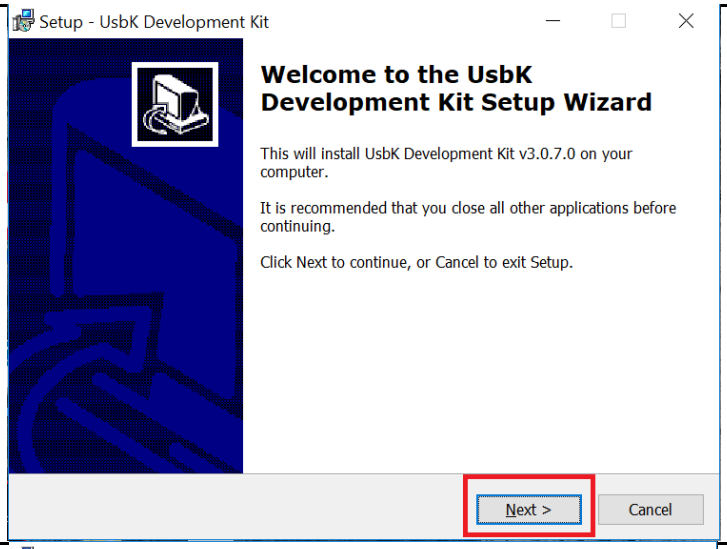
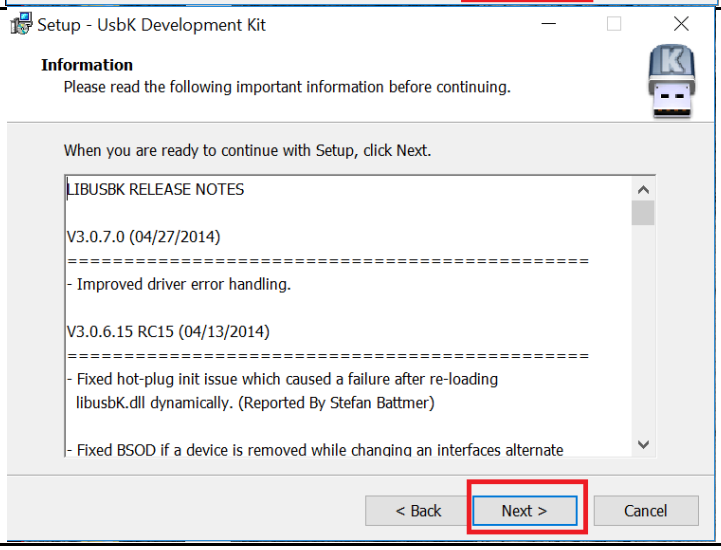
3 LIBUSBK Installation Tutorial

Connect the module to a PC with Windows 10 system. Click libusbK-3.0.7.0-setup.exe to install LIBUSBK driver. Follow below procedure until the driver installed.
Notes: When modules of different VID&PID communicate with the GUI, the USBLIBK driver needs to be installed again.

		Click Next
		Click Next

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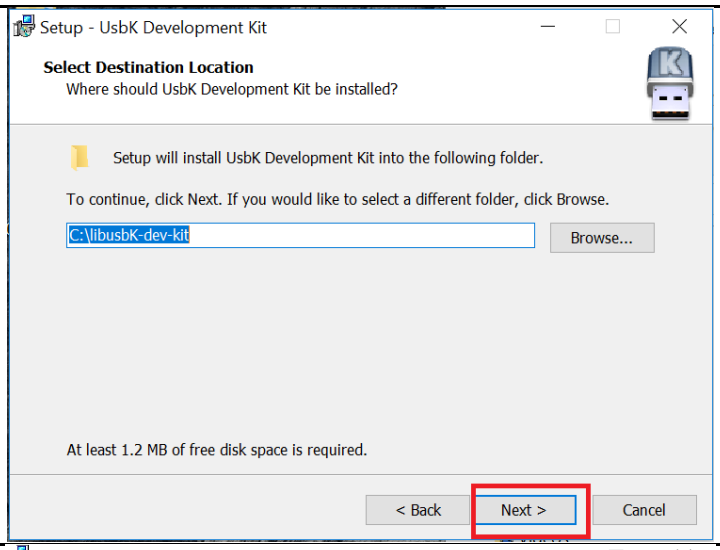
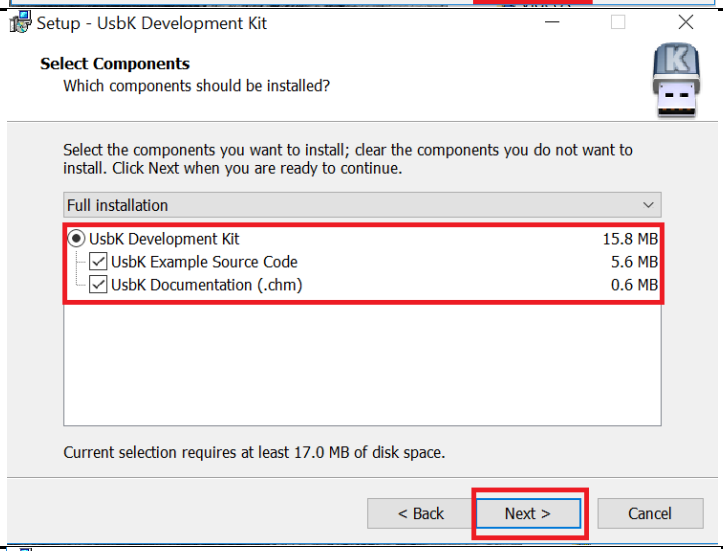
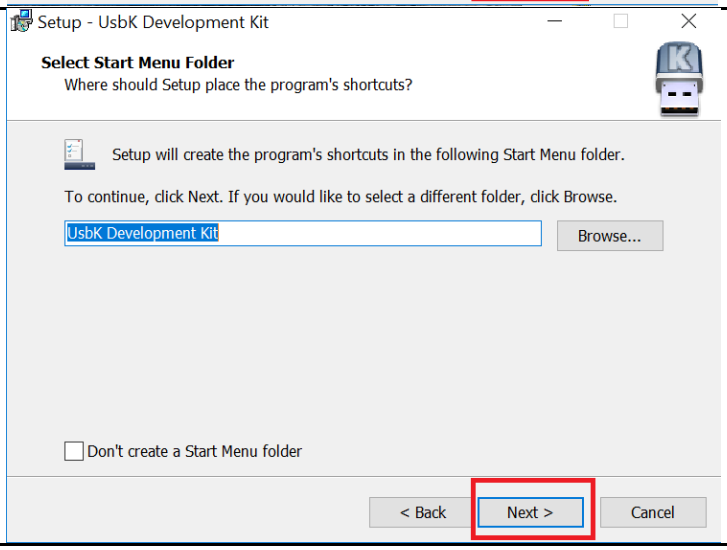
	Choose your folder， and the Click Next
	Select all, and the click next
	Click Next

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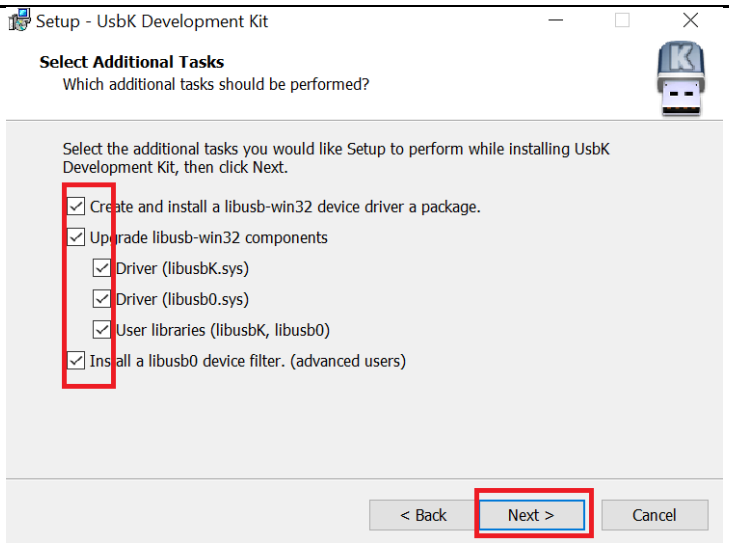
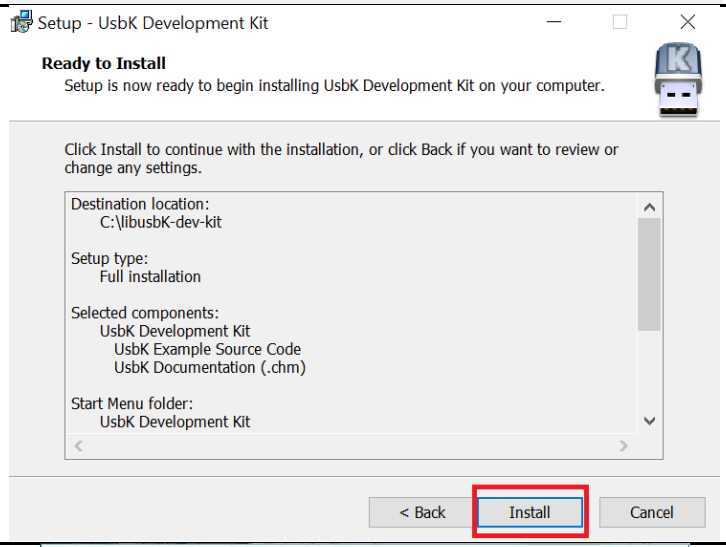
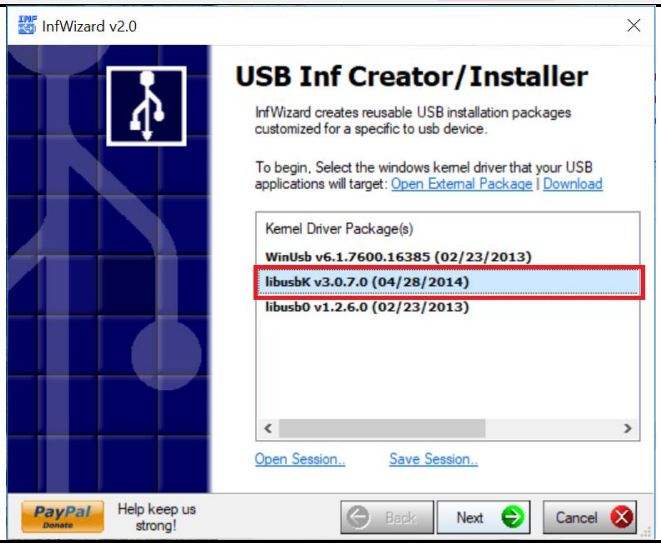
	Select all, and the click next
	Click Install.
	Choose libusbK v3.0.7.0, And then click Next

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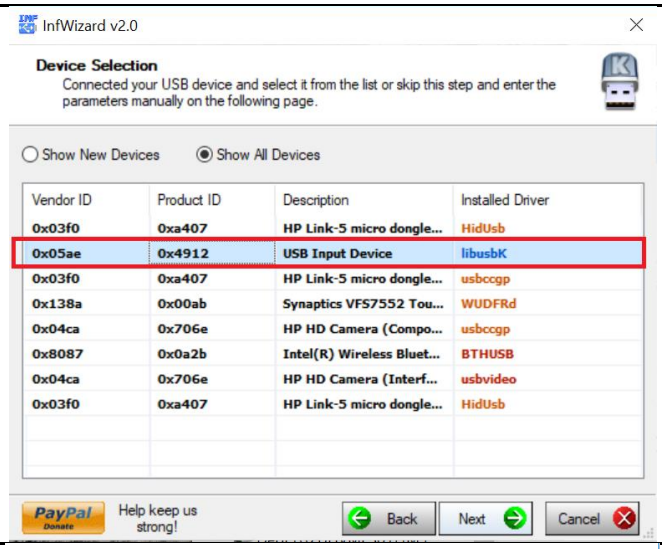
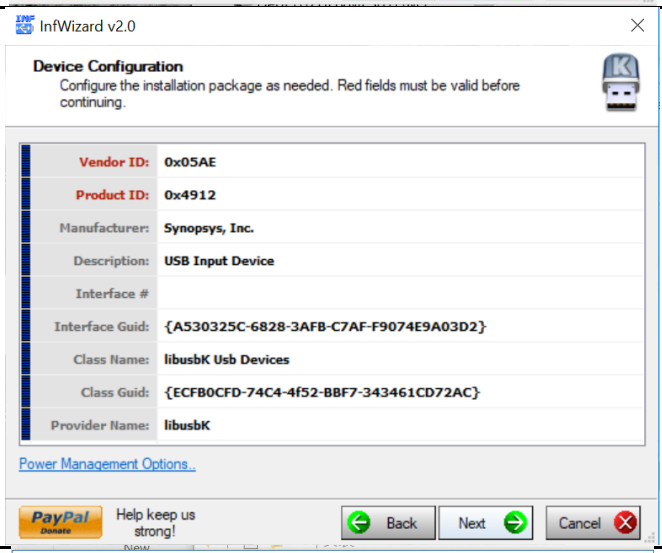
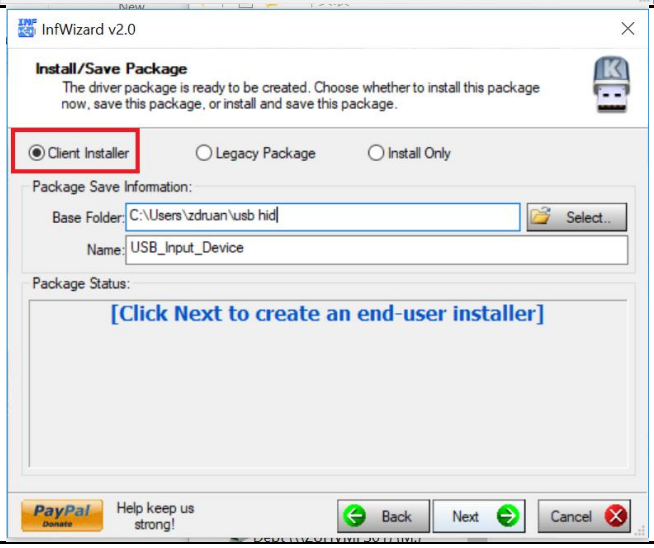


Choose “Show All Device” Select “0x5e 0x4912” and then click Next.



Click Next



Choose Client Installer, choose base folder, Click Next

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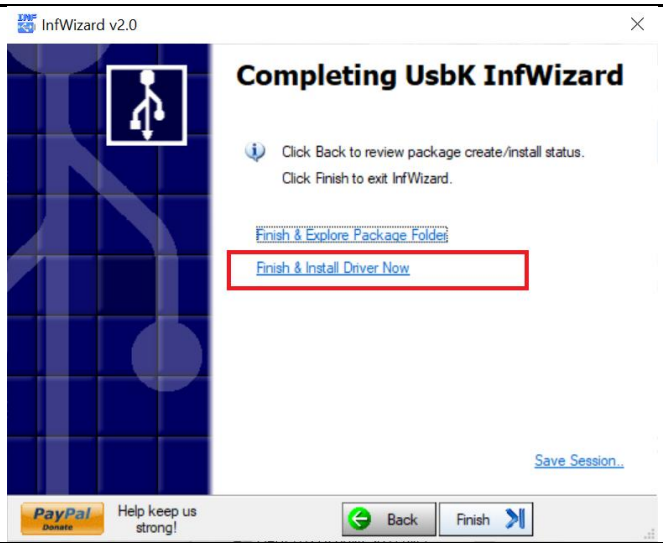
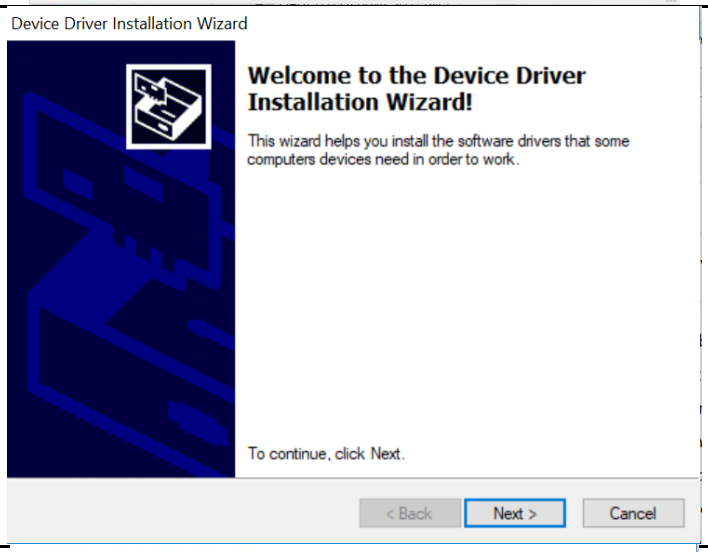
	Choose Finish& Install Driver Now
	Click Next.
	Click Finish.

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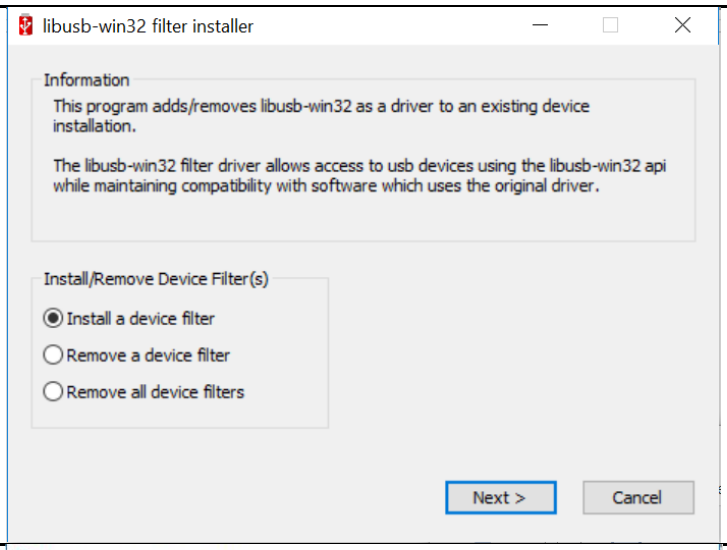
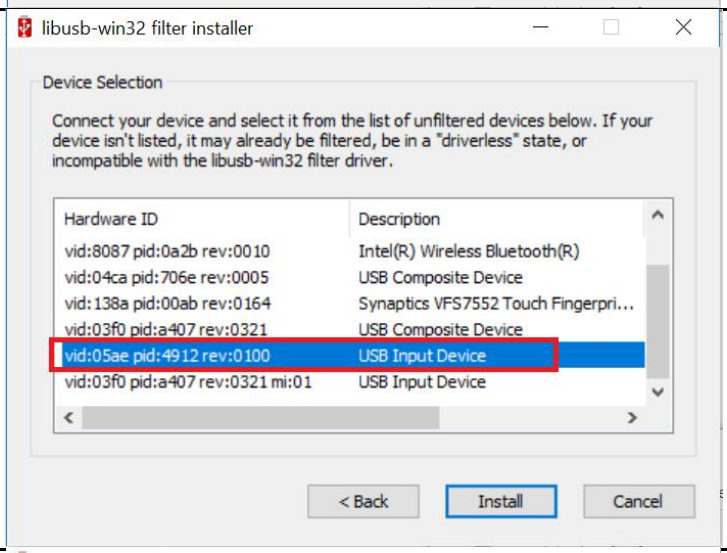
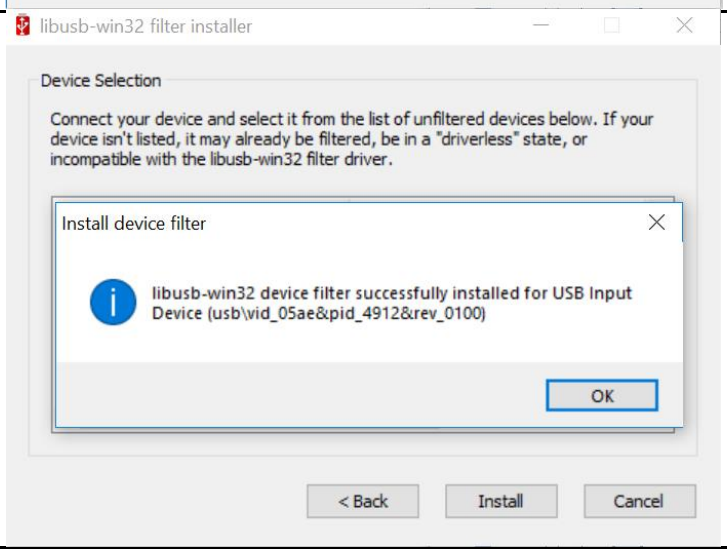
		Choose Install a device filter->click Next
		Choose vid:05ae pid 4912->Click Install Note: if you can't find "vid 05ae, pid:4912 here. Please check if the module is connected. Then, click back and select remove a device filter to check if this PID:VID has already been installed. If yes, you can remove and install it again. PID:VID refer to detailed product type.
		Click OK

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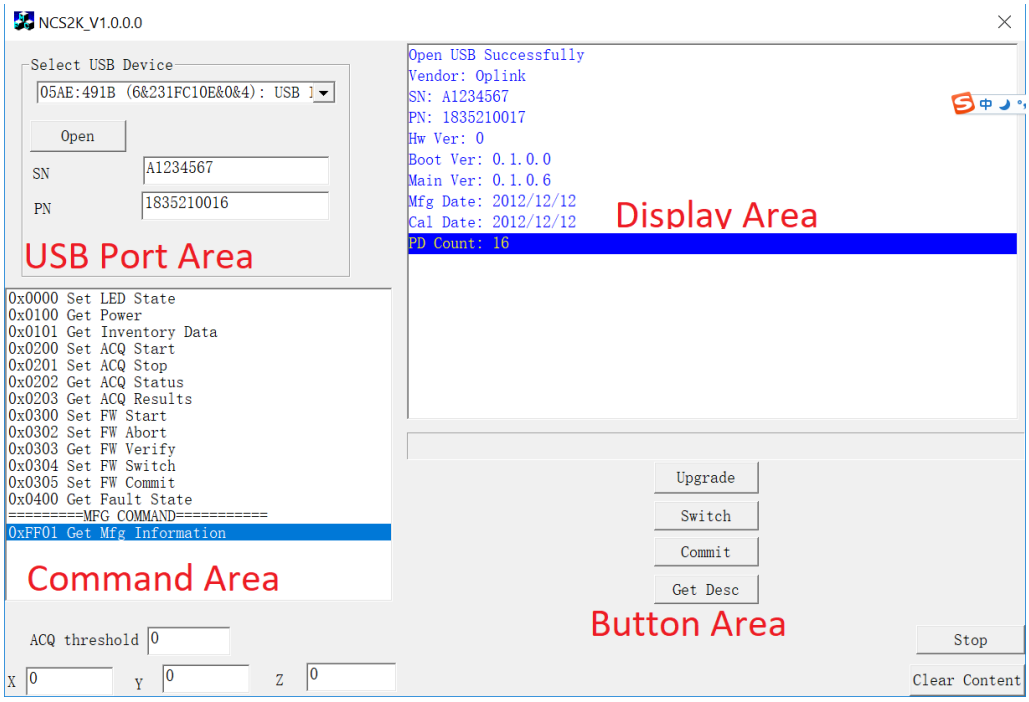
4 GUI Instructions

After LIBUSBK is installed. Click **NCS2K.exe** to execute the GUI. Figure 4.1 shows the interface of the GUI.

Figure 4.1: GUI Interface



There are 4 areas in this GUI.

- USB port Area: Use to connect USB module
- Command Area: All command list here, click them to execute the related command
- Display Area: Display the result of command executing
- Button Area: Button use to control the module.

4.1 Connect Module

Connect the module, select the correct USB VID and PID Port， Click **Open Button**, the result will show on **Display Area. Open USB Successfully** means the module is connected successfully. If the module can't display on the combo box after LIBUSBK was installed successfully, close this GUI and re-open it again. Caution: all the operation can't be executed before this step is successful. If the module is reset or reconnected, you must click Open button to connect.

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4.2 Command Executing

All command related to the module are listed in LIST box. User can execute them by Click it.

Command	Description
Set LED Statue	The command is to control the LED Status. Change the X, Y, Z text and then click this command. The LED will be set to related state. X 1 Y 0 Z 0 X-Green LED, Y-Red LED, Z-Blue LED 0: LED OFF, 1: LED ON, 2: LED Blink 0.5Hz, 4: LED Blink 2.5Hz
Get Power	Retrieve ALL PD latest readings
Get Inventory Data	Retrieve manufacturing information from the module
Set ACQ Start	The command is to start an acquisition The ACQ is Start acquisition threshold, the number was magnified 10 times (eg. -30dBm, we should Fill in -300 in the edit box "ACQ threshold") ACQ threshold -300
Set ACQ Stop	The command is to abort an acquisition
Get ACQ Status	The command is to retrieve the status of the acquisition from the module
Get ACQ Result	The command is to get the acquisition result from the module
Set FW Start	The command is to start FW download process
Set FW Abort	The command is to inform that running download is aborted
Set FW Verify	The command is to verify downloaded code data
Set FW Switch	The command is to switch execution from ACTIVE code to STAND-BY code data.
Set FW Commit	The command is to set running code partition as ACTIVE one
Get Fault State	The command is to retrieve the fault status of the module components after last boot.
MFG Command	
Get MFG Information	Get MFG Information, including firmware version

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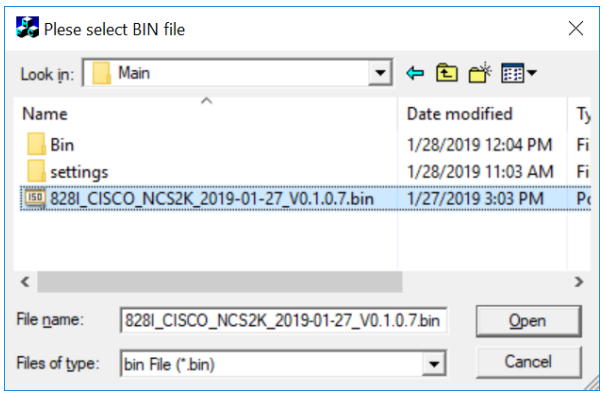
4.3 Firmware Upgrade

After the module is connected, user can follow below procedure to do firmware upgrade.

a. Click **Get MFG Information** in command area to get current firmware version. It can get from Main Ver: (0.1.0.6 for the P0 module) item in the display area.

b. Click **Upgrade button**,



select the upgrade file 828I_CISCO_NCS2K_2019-01-27_V0.1.0.7.bin to do firmware upgrade. Wait until "get firmware verify successfully, 0x0000" appears on the display area.

c. Click **Switch Button**, check if the firmware version is updated. The firmware version should be "Main Ver: 0.1.0.7".

d. If the switch command is successful and the version is "Main Ver: 0.1.0.7", click **Commit Button** to commit this firmware version as default version.

e. Power On/Off the module, click **Get MFG Information** in command area again, check if current firmware version is Main Ver: 0.1.0.7.

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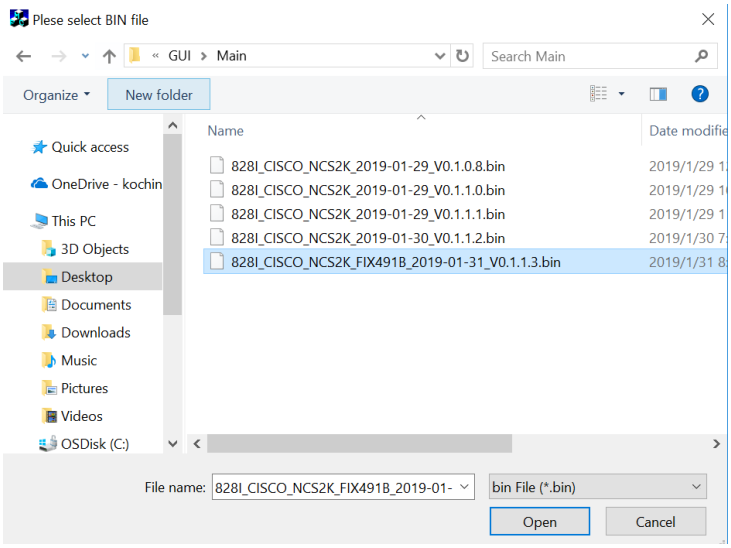
4.4 Upgrade firmware to change Equipment ID (4912 to 491B)

After the module is connected, user can follow below procedure to change the module from 0x4912 to 0x491B.

a. Click **Get MFG Information** in command area to get current firmware version.

b. Click **Upgrade button**,



select the upgrade file 828I_CISCO_NCS2K_FIX491B_2019-01-31_V0.1.1.3.bin to do firmware upgrade. Wait until "get firmware verify successfully, 0x0000" appears on the display area.

c. Click **Switch Button**, below message will display on display area. Don't care it. It is caused by changing of Equipment ID.

```
Set firmware switch successfully
Open USB failed
Get inventory data failed
```

Close the **NCS2K.exe** window, and then open **NCS2K.exe** again, Check if **05AE:491B** shown on Select USB device Combo. If yes, Click **Open** to connect the module. Use **Get Mfg Information** to check if current firmware version is V0.1.1.3. If yes, click **Commit Button** to commit this firmware version as default version.

e. Power On/Off the module, click **Get MFG Information** in command area again, check if current firmware version is Main Ver: V0.1.1.3.

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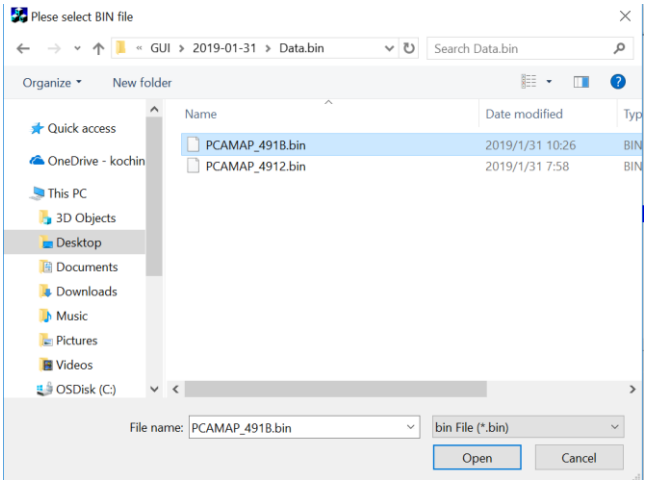
4.5 How to Upgrade PCAMAP bin file

After the module is connected, user can follow below procedure to upgrade binary files. Caution: download binary file will damage the firmware version at backup area. After operation, the switch command will be failed for invalid backup firmware image. User can do firmware upgrade to fix it.

a. Click **Upgrade Button**,



select the upgrade file PCAMAP_xxxx.bin file, wait until "get firmware verify successfully, 0x0000" appears on the display area.

b. **Power On/Off** the module, click **Open Button** to connect the module again.

c. Click **Get Inventory data** to check if the data is as below (0x491B or 0x4912 Module)

```
get firmware verify successfully, 0x0000
0x0000: 04 FF 40 49 1B C0 46 00 4A 01 9E 6D 01 41 01 00
0x0010: 42 41 30 C1 8B 49 49 46 32 31 30 31 30 30 33 54
0x0020: D0 8B 4D 46 2D 38 58 31 30 47 2D 46 4F C6 8A 57
0x0030: 4F 43 55 41 39 52 52 41 41 CB 91 4E 43 53 32 4B
0x0040: 2D 4D 46 2D 38 58 31 30 47 2D 46 4F 89 56 30 31
0x0050: 20 50 01 02 E0 08 01 01 00 01 05 01 00 35 E0 08
0x0060: 01 02 00 02 05 01 00 47 E0 08 01 03 00 03 05 01
0x0070: 00 44 E0 08 01 04 00 04 05 01 00 32 E0 08 09 01
0x0080: 00 09 05 01 00 24 E0 08 09 02 00 0A 08 01 00 20
0x0090: E0 08 09 03 00 0B 08 01 00 39 E0 08 09 04 00 0C
0x00A0: 08 01 00 1D E0 08 08 05 00 01 01 01 00 5B E0 08
0x00B0: 07 05 00 01 02 01 00 49 E0 08 06 05 00 01 03 01
0x00C0: 00 38 E0 08 05 05 00 01 04 01 00 31 E0 08 10 05
0x00D0: 00 09 01 01 00 2D E0 08 0F 05 00 09 02 01 00 34
0x00E0: E0 08 0E 05 00 09 03 01 00 2B E0 08 0D 05 00 09
0x00F0: 04 01 00 30 8C 00 01 09 08 FF 00 00 00 00 00 00
```

PCAMAP of 0x491B

```
0x0000: 04 FF 40 49 12 C0 46 00 4A 01 97 81 01 41 01 00
0x0010: 42 41 30 C1 8B 4F 50 4D 32 33 30 31 31 30 31 D0
0x0020: 8B 46 2D 4D 31 36 4C 43 2D 43 56 8A 57 4F 54 52
0x0030: 44 57 4B 42 41 41 CB 91 4E 43 53 32 4B 2D 4D 46
0x0040: 2D 4D 31 36 4C 43 2D 43 56 89 56 30 31 00 00 00
0x0050: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x0060: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x0070: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x0080: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x0090: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x00A0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x00B0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x00C0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x00D0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x00E0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
0x00F0: FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF
```

PCAMPA of 0x4912

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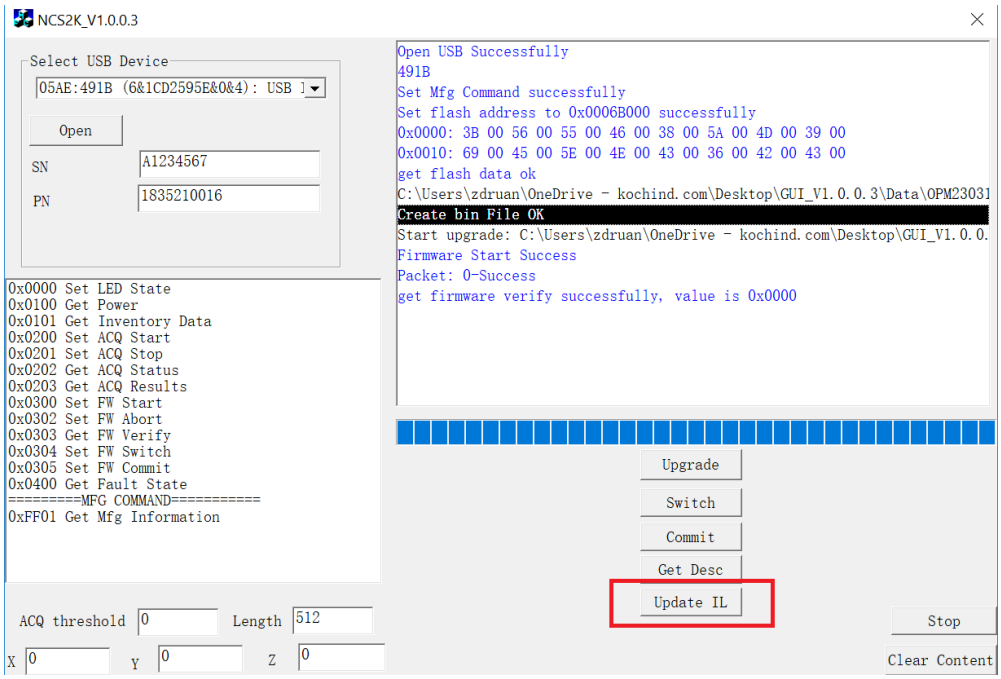
4.6 How to Refresh IL data in PCAMAP

After the module is connected, user can follow below procedure to update IL data to refresh PCAMAP.

Caution: download binary file will damage the firmware version at backup area. After operation, the switch command will be failed for invalid backup firmware image. User can do firmware upgrade to fix it.

a. Click **Update IL Button**, until the display area shows that "get firmware verify successfully, value is 0x0000". (Don't care other text)



b. **Power On/Off** the module, click **Open Button** to connect the module again.

c. Click **Get Inventory data** to check if the data is as below (Just example)

491B_OPM2303
1024.txt

Notes: The Insertion Loss Data format is XXxx(HEX), XX is the IL integer part, xx is IL decimal part(minimum resolution is 0.01).

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